

Introduction to Quantum Computing

QML

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1 Overview

now that we have the math from lecture 1, lets actually build quantum circuits. this lecture covers all the fundamental quantum gates, the circuit model of computation, and two of the most mind-blowing quantum protocols — teleportation and superdense coding.

the goal is to get comfortable with the “building blocks” of quantum computing: the gates u use to construct algorithms, how measurement works in diff bases, and why entanglement is a resource u can exploit.

2 Single Qubit Gates

2.1 The Pauli Gates

the pauli matrices r the most fundamental single qubit gates. they correspond to 180-degree rotations around the x, y, and z axes of the bloch sphere.

Definition 2.1. the Pauli gates:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

along with the identity:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

X gate (quantum NOT / bit flip):

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle$$

$$|\psi\rangle \longrightarrow \boxed{X} \longrightarrow X|\psi\rangle$$

Z gate (phase flip):

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

$$|\psi\rangle \longrightarrow \boxed{Z} \longrightarrow Z|\psi\rangle$$

Y gate:

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle$$

$$|\psi\rangle \longrightarrow \boxed{Y} \longrightarrow Y|\psi\rangle$$

Key Point. key properties of pauli matrices:

- hermitian: $X^\dagger = X$, $Y^\dagger = Y$, $Z^\dagger = Z$
- unitary: $XX^\dagger = I$ (and same for Y, Z)
- self-inverse: $X^2 = Y^2 = Z^2 = I$

- they anticommute: $XY = -YX$, $XZ = -ZX$, $YZ = -ZY$
- $XY = iZ$, $YZ = iX$, $ZX = iY$ (cyclic)
- eigenvalues: always $+1$ and -1
- $\{I, X, Y, Z\}$ forms a basis for all 2×2 matrices

2.2 The Hadamard Gate

Definition 2.2. the Hadamard gate:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

$$|0\rangle \text{ --- } \boxed{H} \text{ --- } |+\rangle$$

the hadamard creates superposition from computational basis states. its probably the most used gate in quantum computing.

Intuition. geometrically, H is a rotation abt the axis halfway between X and Z on the bloch sphere. it maps the Z eigenstates to the X eigenstates and vice versa. $H^2 = I$, so applying it twice gets u back to where u started.

Key Point. the hadamard satisfies:

- $H^2 = I$ (self-inverse)
- $H = \frac{X+Z}{\sqrt{2}}$
- $HXH = Z$ and $HZH = X$ (conjugation swaps X and Z)
- $H|x\rangle = \frac{1}{\sqrt{2}} \sum_{y \in \{0,1\}} (-1)^{xy} |y\rangle$ for $x \in \{0,1\}$

2.3 Phase Gates

Definition 2.3. the S gate (phase gate, $\sqrt{Z}$):

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$S|0\rangle = |0\rangle$, $S|1\rangle = i|1\rangle$. note $S^2 = Z$.

Definition 2.4. the T gate ($\pi/8$ gate, $\sqrt{S}$):

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$T|0\rangle = |0\rangle$, $T|1\rangle = e^{i\pi/4}|1\rangle$. note $T^2 = S$.

$$|\psi\rangle \text{ --- } \boxed{S} \text{ --- } S|\psi\rangle \qquad |\psi\rangle \text{ --- } \boxed{T} \text{ --- } T|\psi\rangle$$

more generally, the general phase gate:

$$R_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

so $S = R_{\pi/2}$, $T = R_{\pi/4}$, and $Z = R_\pi$.

2.4 Rotation Gates

Definition 2.5. the rotation gates parameterized by angle θ :

$$R_x(\theta) = e^{-i\theta X/2} = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (1)$$

$$R_y(\theta) = e^{-i\theta Y/2} = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (2)$$

$$R_z(\theta) = e^{-i\theta Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (3)$$

$$|\psi\rangle \text{ --- } \boxed{R_x(\theta)} \text{ ---} \qquad |\psi\rangle \text{ --- } \boxed{R_y(\theta)} \text{ ---} \qquad |\psi\rangle \text{ --- } \boxed{R_z(\theta)} \text{ ---}$$

these rotation gates r super important for parameterized quantum circuits (PQCs) which r the backbone of variational quantum algorithms and quantum machine learning. well see way more of these in lectures 9+.

special cases:

- $R_x(\pi) = -iX$ (X gate up to global phase)
- $R_y(\pi) = -iY$
- $R_z(\pi) = -iZ$
- $R_z(\pi/2) = \sqrt{Z} \cdot e^{-i\pi/4}$ (S gate up to global phase)

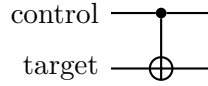
3 Multi-Qubit Gates

3.1 CNOT (Controlled-NOT)

Definition 3.1. the CNOT gate (controlled-X) flips the target qubit if the control qubit is $|1\rangle$:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

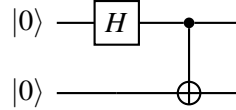
$$\text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle, \quad \text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle$$



in short: $\text{CNOT}|a,b\rangle = |a, a \oplus b\rangle$ where $\oplus$ is addition mod 2.

Key Point. CNOT is the standard entangling gate. u cant create entanglement with single-qubit gates alone — u need at least one two-qubit gate, and CNOT is the most common choice.

Example 3.1. creating a bell state with H and CNOT:



step by step:

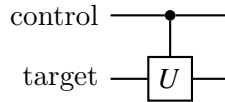
$$|00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \quad (4)$$

$$\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle \quad (5)$$

3.2 Controlled Gates in General

Definition 3.2. a controlled-U gate applies unitary U to the target qubit when the control qubit is $|1\rangle$:

$$C-U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U = \begin{pmatrix} I & 0 \\ 0 & U \end{pmatrix}$$



common controlled gates:

- CX = CNOT (controlled-X)
- CZ (controlled-Z): applies Z to target when control is $|1\rangle$
- CS (controlled-S)
- CRz (controlled rotation)

Key Point. CZ is symmetric — it doesnt matter which qubit is “control” and which is “target”:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

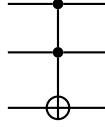
CZ adds a phase of -1 only to $|11\rangle$. both qubits play the same role.

3.3 Toffoli Gate (CCNOT)

Definition 3.3. the Toffoli gate (controlled-controlled-NOT) flips the target qubit only when both control qubits are $|1\rangle$:

$$\text{Toffoli } |a, b, c\rangle = |a, b, c \oplus (a \cdot b)\rangle$$

its an 8×8 matrix that acts as the identity on all basis states except $|110\rangle \leftrightarrow |111\rangle$.



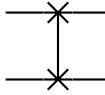
Intuition. the toffoli gate is universal for classical reversible computation. if u set the target qubit to $|0\rangle$, then the output is $|a, b, a \text{ AND } b\rangle$ — its computing a classical AND gate reversibly. combined with the ability to set inputs to constants, u can simulate any classical boolean circuit.

3.4 SWAP Gate

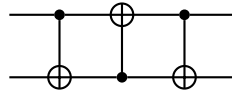
Definition 3.4. the SWAP gate exchanges two qubits:

$$\text{SWAP } |a, b\rangle = |b, a\rangle$$

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



SWAP can be decomposed into three CNOTs:



Example 3.2. verify: apply three CNOTs to $|ab\rangle$:

$$|a, b\rangle \xrightarrow{\text{CNOT}_{1 \rightarrow 2}} |a, a \oplus b\rangle \quad (6)$$

$$\xrightarrow{\text{CNOT}_{2 \rightarrow 1}} |a \oplus (a \oplus b), a \oplus b\rangle = |b, a \oplus b\rangle \quad (7)$$

$$\xrightarrow{\text{CNOT}_{1 \rightarrow 2}} |b, b \oplus (a \oplus b)\rangle = |b, a\rangle \quad (8)$$

the qubits are swapped.

4 Universal Gate Sets

Definition 4.1. a set of gates is universal if any unitary operation can be approximated to arbitrary precision using gates from the set.

Key Point. common universal gate sets:

- $\{H, T, \text{CNOT}\}$ — the standard choice. any quantum computation can be built from these three gates (plus their inverses)
- $\{H, S, T, \text{CNOT}\}$ — also universal (S is redundant but convenient)
- $\{R_y(\theta), R_z(\theta), \text{CNOT}\}$ for all θ — continuous universal gate set
- any entangling two-qubit gate + all single-qubit gates is universal

Intuition. the solovay-kitaev theorem says that any single-qubit gate can be approximated to precision ϵ using $O(\log^c(1/\epsilon))$ gates from a finite universal set (like $\{H, T\}$). this is important bcs it means we dont need infinitely precise gate angles — a finite gate set suffices. in practice, quantum hardware implements a fixed set of native gates, and the quantum compiler decomposes arbitrary unitaries into these native gates.

Example 4.1. decomposing the S gate into H and T gates:

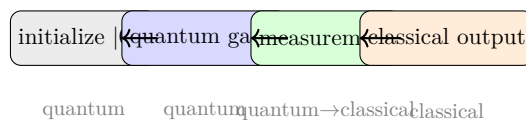
recall $S = T^2$. so applying T twice gives u S. alternatively:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}^2$$

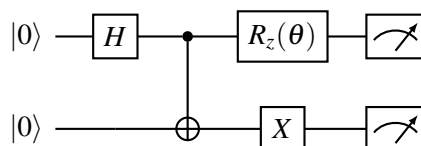
5 The Quantum Circuit Model

5.1 How to Read a Quantum Circuit

the quantum circuit model is the standard computational framework:



quantum circuits r read left to right. each horizontal line is a qubit. gates r applied in the order they appear from left to right. time flows left to right.



this circuit:

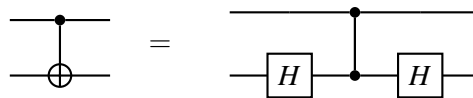
1. starts with two qubits in $|00\rangle$
2. applies H to qubit 0
3. applies CNOT with qubit 0 as control, qubit 1 as target
4. applies $R_z(\theta)$ to qubit 0 and X to qubit 1
5. measures both qubits

5.2 Circuit Identities

some useful identities for simplifying circuits:

- $HXH = Z$ (conjugating X by H gives Z)
- $HZH = X$ (conjugating Z by H gives X)
- $HYH = -Y$
- $X = HZH$ (can replace X with H-Z-H)
- $\text{CNOT}_{12} = (I \otimes H) \cdot \text{CZ}_{12} \cdot (I \otimes H)$

Example 5.1. the following two circuits are equivalent:

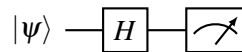


bcs $\text{CNOT} = (I \otimes H) \cdot \text{CZ} \cdot (I \otimes H)$ and CZ is symmetric.

6 Measurement in Different Bases

6.1 Measuring in the X Basis

to measure in the $\{|+\rangle, |-\rangle\}$ basis, apply H before measuring in the computational basis:



Intuition. applying H before measurement rotates the measurement basis. instead of projecting onto $|0\rangle$ and $|1\rangle$, we are effectively projecting onto $|+\rangle$ and $|-\rangle$. outcome 0 means the state was closer to $|+\rangle$, outcome 1 means closer to $|-\rangle$.

Example 6.1. measure $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$ in the X basis:

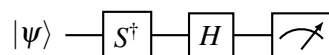
apply H: $H|+\rangle = |0\rangle$.

measure in computational basis: get 0 with probability 1.

so measuring $|+\rangle$ in the X basis always gives outcome “+”. makes sense.

6.2 Measuring in the Y Basis

to measure in the Y basis $\{|+i\rangle, |-i\rangle\}$, apply $S^\dagger H$ before measuring:



6.3 General Measurement Basis

to measure in any orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$, find the unitary U that maps $|b_i\rangle \rightarrow |i\rangle$, apply U , and measure in the computational basis.

7 The No-Cloning Theorem

Definition 7.1. the no-cloning theorem states that there is no quantum operation that can copy an arbitrary unknown quantum state:

$$\nexists U \text{ such that } U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \text{ for all } |\psi\rangle$$

proof: suppose such a U exists. then for any two states $|\phi\rangle$ and $|\psi\rangle$:

$$U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle \quad (9)$$

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad (10)$$

take the inner product of both sides:

$$\begin{aligned} \langle\phi|\psi\rangle\langle 0|0\rangle &= \langle\phi|\psi\rangle\langle\phi|\psi\rangle \\ \langle\phi|\psi\rangle &= \langle\phi|\psi\rangle^2 \end{aligned}$$

this equation has only two solutions: $\langle\phi|\psi\rangle = 0$ or $\langle\phi|\psi\rangle = 1$. meaning either the states are identical or orthogonal. u cant clone arbitrary states — contradiction. $\square$

Key Point. no-cloning has huge implications:

- u cant copy quantum information — this is why quantum error correction is fundamentally different from classical error correction (cant just duplicate the data)
- u cant measure a state without disturbing it — bcs measurement “collapses” the state
- quantum cryptography (QKD) works bcs eavesdroppers cant copy quantum signals without being detected
- in QML, u cant simply “look at” intermediate quantum states during computation without affecting the result

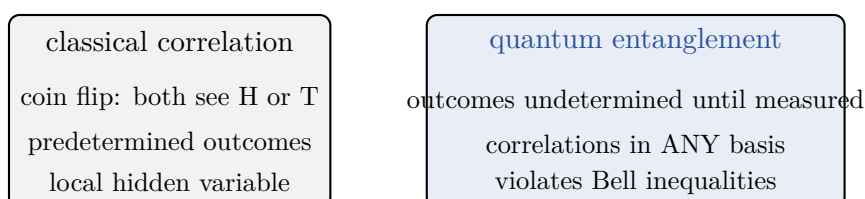
8 Entanglement

8.1 What Entanglement Actually Is

we introduced entanglement in lecture 1 using the tensor product factorization test. now lets go deeper.

Definition 8.1. a pure state $|\psi_{AB}\rangle$ of a bipartite system is entangled if it cannot be written as a product state $|a\rangle \otimes |b\rangle$. equivalently, a state is entangled iff the reduced density matrix of either subsystem is mixed (has purity < 1).

Intuition. entanglement means the whole system has more information than the sum of its parts. if u know everything abt the global state (its pure), but the local states are mixed (uncertain), then the information is encoded in correlations between the subsystems. these correlations are stronger than any classical correlations can be — thats the essence of entanglement.



8.2 Bell States

the four bell states r the maximally entangled two-qubit states:

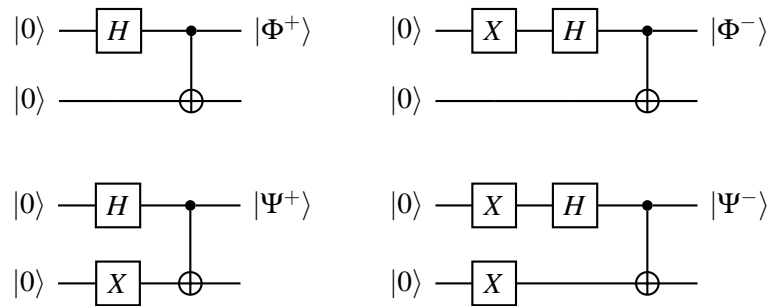
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (11)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \quad (12)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (13)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (14)$$

circuits to create all four bell states from $|00\rangle$:

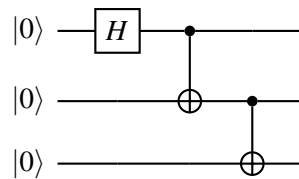


8.3 GHZ States

Definition 8.2. the GHZ state for n qubits:

$$|GHZ_n\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n})$$

circuit for 3-qubit GHZ:



step by step:

$$|000\rangle \xrightarrow{H_1} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|00\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle) \quad (15)$$

$$\xrightarrow{\text{CNOT}_{12}} \frac{1}{\sqrt{2}}(|000\rangle + |110\rangle) \quad (16)$$

$$\xrightarrow{\text{CNOT}_{23}} \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) = |GHZ_3\rangle \quad (17)$$

8.4 Quantifying Entanglement

for pure bipartite states, entanglement can be measured using the von Neumann entropy of the reduced density matrix:

$$S(\rho_A) = -\text{tr}(\rho_A \log_2 \rho_A) = -\sum_i \lambda_i \log_2 \lambda_i$$

where λ_i r the eigenvalues of ρ_A .

- $S = 0$: product state (no entanglement)
- $S = 1$: maximally entangled (for two qubits)
- $0 < S < 1$: partially entangled

Example 8.1. for the bell state $|\Phi^+\rangle$, we showed $\rho_A = I/2$, which has eigenvalues $1/2, 1/2$:

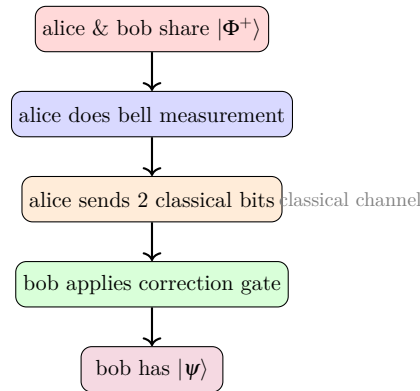
$$S = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = -\frac{1}{2}(-1) - \frac{1}{2}(-1) = 1 \text{ ebit}$$

maximally entangled, as expected.

9 Quantum Teleportation

quantum teleportation is one of the most beautiful protocols in quantum information. it allows u to transmit an unknown quantum state using only classical communication and a pre-shared entangled pair.

heres the high-level protocol as a flowchart:



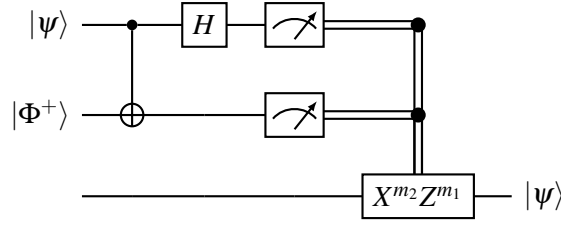
9.1 Setup

alice has an unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that she wants to send to bob. they share a bell pair $|\Phi^+\rangle$.

total initial state (3 qubits: alice's data qubit, alice's half of bell pair, bob's half of bell pair):

$$|\psi\rangle_1 \otimes |\Phi^+\rangle_{23} = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

9.2 The Protocol



9.3 Step-by-Step Walkthrough

expand the initial state:

$$\begin{aligned} & \frac{1}{\sqrt{2}}[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)] \\ &= \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle] \end{aligned}$$

step 1: CNOT on qubits 1,2:

$$\frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]$$

step 2: H on qubit 1:

using $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$\frac{1}{2}[\alpha(|0\rangle + |1\rangle)|00\rangle + \alpha(|0\rangle + |1\rangle)|11\rangle + \beta(|0\rangle - |1\rangle)|10\rangle + \beta(|0\rangle - |1\rangle)|01\rangle] \quad (18)$$

regroup by alice's measurement outcomes (first two qubits):

$$= \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \quad (19)$$

$$+ |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)] \quad (20)$$

step 3: alice measures qubits 1 and 2, gets one of four outcomes with equal probability 1/4:

alice measures	bob's state	correction
$ 00\rangle$	$\alpha 0\rangle + \beta 1\rangle = \psi\rangle$	none (I)
$ 01\rangle$	$\alpha 1\rangle + \beta 0\rangle = X \psi\rangle$	apply X
$ 10\rangle$	$\alpha 0\rangle - \beta 1\rangle = Z \psi\rangle$	apply Z
$ 11\rangle$	$\alpha 1\rangle - \beta 0\rangle = ZX \psi\rangle$	apply ZX

step 4: alice sends her 2 classical bits to bob. bob applies the corresponding correction. result: bob has $|\psi\rangle$.

Key Point. teleportation doesn't violate no-cloning or special relativity:

- alice's original state is destroyed (no cloning) — after measurement, her qubits are in a definite state $|m_1 m_2\rangle$, not $|\psi\rangle$

- bob needs alice’s classical bits to reconstruct the state (no FTL communication) — without the correction, bob’s qubit looks maximally mixed
- the state is “teleported” not “transmitted” — no physical qubit travels from alice to bob

Intuition. teleportation is a resource conversion protocol. it converts 1 ebit of entanglement + 2 classical bits into 1 qubit of quantum communication. the entanglement acts as a “quantum channel” that was established beforehand.

10 Superdense Coding

superdense coding is the “reverse” of teleportation. instead of sending a quantum state using classical communication + entanglement, u send 2 classical bits by transmitting only 1 qubit (plus using 1 ebit of pre-shared entanglement).

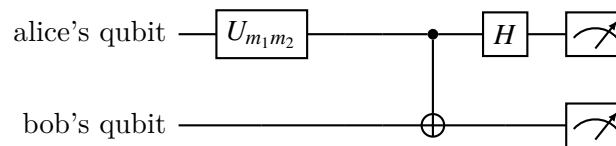
10.1 The Protocol

alice and bob share a bell pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

alice wants to send 2 classical bits m_1m_2 to bob. she applies a gate to her qubit based on the message:

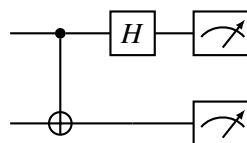
message	alice applies	resulting state
00	I	$ \Phi^+\rangle = \frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$
01	X	$ \Psi^+\rangle = \frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$
10	Z	$ \Phi^-\rangle = \frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$
11	ZX	$ \Psi^-\rangle = \frac{1}{\sqrt{2}}(10\rangle - 01\rangle)$

alice sends her qubit to bob. now bob has both qubits of a bell state. he performs a bell measurement (CNOT then H, then measure) to determine which bell state it is, recovering the 2-bit message.



10.2 Bell Measurement

the key circuit for distinguishing bell states:



this maps:

$$|\Phi^+\rangle \rightarrow |00\rangle \quad (21)$$

$$|\Psi^+\rangle \rightarrow |01\rangle \quad (22)$$

$$|\Phi^-\rangle \rightarrow |10\rangle \quad (23)$$

$$|\Psi^-\rangle \rightarrow |11\rangle \quad (24)$$

Example 10.1. verify for $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

$$\text{CNOT: } \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = |+\rangle|0\rangle$$

$$\text{H on first qubit: } H|+\rangle = |0\rangle$$

final state: $|00\rangle$. measurement gives 00.

Key Point. comparison of teleportation and superdense coding:

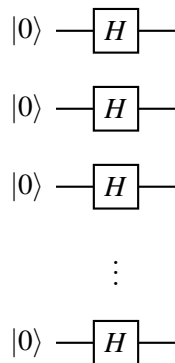
	teleportation	superdense coding
input	1 qubit	2 classical bits
resource	1 ebit + 2 cbits	1 ebit + 1 qubit
output	1 qubit (at bob)	2 classical bits (at bob)

they r “dual” protocols — one converts ebits + cbits into qubits, the other converts ebits + qubits into cbits.

11 Important Quantum Circuit Patterns

11.1 Uniform Superposition

to create an equal superposition over all n -qubit basis states:



$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

this creates a uniform superposition over all 2^n computational basis states using just n hadamard gates. its the starting point for many quantum algorithms.

11.2 Phase Kickback

Key Point. phase kickback is a crucial phenomenon where a controlled-U gate “kicks back” the eigenvalue phase onto the control qubit.

if $U|u\rangle = e^{i\phi}|u\rangle$ (i.e., $|u\rangle$ is an eigenstate of U with eigenvalue $e^{i\phi}$), then:

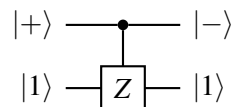
$$C-U|+\rangle|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle|u\rangle + e^{i\phi}|1\rangle|u\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)|u\rangle$$

the phase $e^{i\phi}$ appears on the control qubit, not the target. the target is unchanged.

phase kickback is the mechanism behind:

- the deutsch-jozsa algorithm
- quantum phase estimation
- grover’s diffusion operator
- many other quantum algorithms

Example 11.1. phase kickback with the Z gate. since $Z|1\rangle = -|1\rangle$ ($|1\rangle$ is an eigenstate of Z with eigenvalue -1):



the control qubit picks up the phase: $|+\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + (-1)|1\rangle) = |-\rangle$.

the target stays as $|1\rangle$.

12 Summary

Key Point. wt we covered:

1. single qubit gates: pauli gates (X, Y, Z), hadamard (H), phase gates (S, T), rotation gates (R_x , R_y , R_z)
2. multi-qubit gates: CNOT, CZ, toffoli, SWAP
3. universal gate sets: $\{H, T, \text{CNOT}\}$ is enough for any quantum computation
4. circuit model: circuits r read left to right, gates r matrices, composition is matrix multiplication
5. measurement: can measure in any basis by applying a basis-changing unitary first
6. no-cloning: u cant copy unknown quantum states — fundamental constraint
7. entanglement: bell states, GHZ states, quantified by von neumann entropy
8. teleportation: send a qubit using 1 ebit + 2 cbits

9. superdense coding: send 2 cbits using 1 ebit + 1 qubit
10. phase kickback: eigenvalue phases “kick back” onto control qubits — key mechanism for many algorithms

these are the building blocks. every quantum algorithm, including QML algorithms, is built from these components.